

US Technology

Key takeaways from our Silicon Valley tech tour

Equities Semiconductors & Equipment

United States

- ◆ Key takeaways from our meetings with 27 tech companies in Silicon Valley along with a group of investors
- ◆ No slowdown in demand as supply chain bottlenecks are expected to continue through 2027 or possibly longer
- ◆ Monetization of AI remains a key debate; AI exposure led re-rating unlikely with focus now on earnings upside potential

We held meetings with 27 tech companies in Silicon Valley, along with a group of investors, across five days. We highlight the key takeaways from these meetings in this note.

Supply chain bottlenecks to persist through 2027 at least: There is a broad, AI-led supply crunch rather than a single isolated constraint. Demand continues to run ahead of supply across much of the semiconductor and data center ecosystem. The major bottlenecks are across 1) data center capacity and power; 2) substrates, foundry wafers, and advanced packaging; 3) optical components; and 4) memory. Companies are responding by pre-booking capacity, diversifying suppliers, vertically integrating to the extent possible, and holding more inventory. However, the capacity bottlenecks are expected to persist through 2027, with a possibility of extending beyond that too.

No signs of demand slowdown; focus on open-source models: There are no signs of AI-related demand slowing down. However, demand is broadening from training towards inference, agents, enterprise, sovereign AI, storage, and physical AI. The focus is also shifting towards open-source models due to cost, flexibility, and sovereignty. China's push in open-source models has triggered an urgency to ramp up open-source models in the US. However, both open-source and proprietary / closed-source models are expected to co-exist.

AI monetization remains a key debate: AI monetization remains a key debate among investors with software companies acknowledging that though AI monetization is moving beyond experimentation, they are still unable to quantify the monetization story. The revenue contribution remains modest relative to the underlying usage and investment. Companies that sit inside the workflow or communications layer rather than the ones trying to build frontier models themselves seem to be better set-up for monetization.

Bottom-up earnings upside remains key for re-rating: The demand is expected to remain strong but constrained by supply capacity. While hyperscaler capex momentum remains critical to support growth of semi companies, enterprise and sovereign AI adoption are also becoming more important as they help support a wider set of beneficiaries. With this backdrop, we expect the multiples of semi and tech companies to remain capped with upside driven by bottom-up earnings performance.

Frank Lee*

Global Head of Tech Hardware & Semi Research
The Hongkong and Shanghai Banking Corporation Limited
frank.lee@hsbc.com.hk
+852 2996 6916

Pulkit Aggarwal*

Analyst, Semis and Tech Hardware
HSBC Securities and Capital Markets (India) Private Limited
pulkit.aggarwal@hsbc.co.in
+91 86 5529 3272

* Employed by a non-US affiliate of HSBC Securities (USA) Inc, and is not registered/ qualified pursuant to FINRA regulations

Disclosures & Disclaimer

This report must be read with the disclosures and the analyst certifications in the Disclosure appendix, and with the Disclaimer, which forms part of it.

Issuer of report: The Hongkong and Shanghai Banking Corporation Limited

View HSBC Global Investment Research at:
<https://www.research.hsbc.com>

Ratings and target prices

Company	Ticker	Currency	Current price	TP	Rating	Implied upside	Mkt cap USDm	EPS growth	
								2026e	2027e
NVIDIA	NVDA US	USD	227.38	365.00	Buy	61%	5,479,858	94%	58%
Intel	INTC US	USD	121.78	200.00	Buy	64%	643,616	245%	70%
Marvell	MRVL US	USD	257.38	325.00	Buy	26%	225,696	52%	79%
Qualcomm	QCOM US	USD	194.23	165.00	Hold	-15%	203,941	-13%	-4%
GlobalFoundries Inc.	GFS US	USD	48.57	55.00	Hold	13%	26,653	14%	31%

Source: HSBC estimates, Bloomberg. Priced at close of 21 Sep 2026

Supply chain bottlenecks

As highlighted above, the supply crunch is broad-based and not an isolated constraint. We summarize some of the key areas and major bottlenecks discussed by the companies.

Data center capacity and power

- ◆ **DeepInfra:** DeepInfra can build an inference cluster in as little as 90 days, but securing suitable data center space is becoming increasingly difficult.

The main constraints are:
 1. Limited availability of data center space.
 2. Removal of tax incentives in some US states.
 3. Securing liquid cooled clusters to accommodate next-gen chips
- ◆ **Celesta Capital:** Celesta described major obstacles to building terrestrial data centers in the US. Around 60% of projects are reportedly facing organized protests or political challenges. This creates delays around permitting, power availability, and local acceptance. The firm sees space-based data centers as a possible long-term alternative, although transport and deployment costs are currently around 20 times those of terrestrial facilities.
- ◆ **NVIDIA:** NVIDIA said many older data centers cannot support the power and liquid-cooling requirements of newer systems such as Vera Rubin. As a result, much of current customer spending is effectively greenfield, requiring new infrastructure rather than simple equipment upgrades.

Substrates, foundry wafers, and advanced packaging

- ◆ **NVIDIA:** NVIDIA's demand remains well above available supply. Management referred to next year's growth of 70% as supply-constrained, while saying unconstrained demand is materially higher. The company's ability to secure wafer allocation from TSMC is strategically important.
- ◆ **Intel:** Intel said server demand is strong but that it can currently fulfil only around 50% of potential server demand. The principal constraints are:
 1. ABF substrate availability is more of a constraint than wafer supply
 2. Yield challenges associated with large reticle sizes for EMIB-T packaging
 3. Supplier learning curves for advanced packaging substrates
- ◆ **GlobalFoundries:** GlobalFoundries described industry capacity as tight, particularly for processes linked to data centers.
- ◆ **SoftBank:** SoftBank highlighted continuing foundry tightness and is working with TSMC while exploring other capacity options. Its broader strategy is tied to localizing semiconductor and packaging supply chains, particularly in the US and Japan.

- ◆ **Qualcomm:** The USD5bn FY27 data center revenue target is described as supply-constrained rather than demand-constrained. The company has secured good visibility for FY2027 but still faces tight supply in advanced packaging, substrates, testers, and memory stacking.
- ◆ **Applied materials:** Applied Materials described a more complex manufacturing environment, driven by 1) increasing logic content in DRAM; 2) advanced 300mm packaging; 3) through-silicon via (TSV) and wafer-level packaging; and 4) higher requirements for yield, power efficiency, and variability control.

Optical components

- ◆ **Arista Networks:** Arista is supply-constrained and expects the situation to continue through at least 2027. Demand is exceeding supply, with constraints particularly affecting 3nm chips, transceivers, and high-speed optical components. The company believes that if supply unlocks earlier than expected, it could exceed its current 40% growth guidance.
- ◆ **Lightmatter:** Lightmatter identified structural networking bottlenecks rather than simple component shortages:
 1. Insufficient bandwidth per switch.
 2. The need for longer reach than what is provided by one-meter copper cables.
 3. The physical “shoreline” limitation of conventional chips, where I/O can only be placed around the chip perimeter.

According to Lightmatter, the broader ecosystem is also dealing with delayed CPO adoption, with expected timelines moving towards 2028-30.

- ◆ **Lumentum:** Lumentum described one of the clearest component-level bottlenecks: a severe shortage of 200G-lane continuous-wave lasers. Lumentum also highlighted limited supply of six-inch laser substrates, with some sulphur-doped substrates reportedly achieving yields as low as 5%.
- ◆ **Marvell:** Marvell identified bottlenecks across the optical supply chain, particularly in lasers, silicon photonics and assembly.

Memory

- ◆ **Cohu:** Cohu sees growing demand for HBM inspection because the cost of a failed die in a completed GPU board can reach approximately USD25k. According to Cohu, the bottlenecks are primarily operational like 1) a large number of simultaneous customer qualifications; 2) limited engineering resources; 3) the need to maintain lead times and product availability for existing customers; and 4) increasing complexity in testing HBM3, HBM4 and eventually HBM5
- ◆ **Seagate:** The main supply-chain issue is the long production cycle for heat-assisted magnetic recording (HAMR) components. Heads can take three to four quarters to produce, creating a transition lag between investment and revenue.
- ◆ **Western Digital:** Western Digital identified constraints in NAND and DRAM components used in bill of materials. Inflation and input-cost pressure are making the traditional 10% annual cost-reduction target more difficult to achieve. The company expects to pass some of these costs through while using long-term agreements to support more predictable pricing.

Demand sustainability

The comments from the companies support sustained AI demand with demand broadening beyond training towards inference, agents, enterprise, sovereign AI, and physical AI.

Structural arguments supporting durable demand

- ◆ **Inference and AI agents could materially expand usage:** DeepInfra expects spending to shift from training to inference. AI agents may consume tens of millions of tokens in task loops, particularly for coding, creating much higher usage than human chat. NVIDIA also expects inference eventually to become the larger market.
- ◆ **Demand is moving beyond a small group of hyperscalers:** NVIDIA said growth is broadening into AI clouds, enterprise, and sovereign customers. Its ACIE segment already receives more than half of its revenue from AI clouds and “neo-clouds”, while enterprise and sovereign adoption is still at an earlier stage and could outgrow hyperscale demand.
- ◆ **AI infrastructure demand remains supply-constrained:** Multiple companies acknowledged that demand remains strong but remains supply-constrained capping further growth.
 - **Arista Networks:** demand exceeds supply, with constraints expected through at least 2027.
 - **Intel:** able to fulfil only around 50% of potential server demand, with substrates a key constraint.
 - **Qualcomm:** USD5bn data center target is supply-constrained rather than demand-constrained
- ◆ **AI is creating new infrastructure requirements, not simply replacing existing workloads:** Seagate argued that AI agents, longer context windows, and physical AI will drive more storage for video, logs, sensor data, and compliance records. Hard disk drives (HDDs) remain economically attractive because flash can cost 20-25 times more per terabyte.
- ◆ **Enterprise and sovereign AI could provide a more durable demand base:** HP Enterprise sees enterprise and sovereign deployments as less exposed to hyperscaler capex cycles and less vulnerable to white-box commoditization. Its Private Cloud AI offering is positioned as delivering 30-60% cost savings versus public-cloud, token-based models.

Evidence of confidence

- ◆ **SoftBank** dismissed AI bubble concerns, saying demand for computing power remains well above supply.
- ◆ **Monolithic Power Systems** described AI/data center demand as healthy and increasingly diversified, with ASIC revenue likely already exceeding GPU revenue.
- ◆ **GlobalFoundries** reported that recent silicon-photonics demand signals were “flashing green lights”, with more than 40 active customer engagements and a target of over USD1bn in silicon-photonics revenue by 2028.
- ◆ **Intel** believes its addressable market is now around 15x larger than historical, supporting confidence in sustained double-digit growth.
- ◆ **Cohu** noted customer equipment utilization has reached the 80% threshold at which customers historically begin buying additional capacity.

Areas of caution

- ◆ **Demand sustainability depends on economics:** Companies repeatedly referenced power, cooling, data center capacity, financing costs, and supply availability as constraints.
- ◆ **Some demand is still greenfield and dependent on infrastructure build-out:** NVIDIA said much current customer spending is greenfield because older data centers cannot support the power and liquid-cooling requirements of newer systems. This supports short-term demand but also makes timing dependent on construction, energy, and supply-chain execution.

Focus shift towards open-source models

The discussion favors open-source or open weight models for cost, flexibility, and sovereignty, while proprietary/closed-source models retain advantages where frontier performance, safety, workflow integration, or enterprise-grade capabilities are critical. Previously the US companies were more focused on frontier models. However, since our Silicon Valley Tech Tour in June, things seem to have changed, especially after the Chinese open-source models reset industry expectations about compute price. As a result, open-source models have become a key debate in the US too.

Cost advantages of open-source models

- ◆ **DeepInfra** focuses on Llama, Mistral, DeepSeek, and GLM. It said these models can run around ten times cheaper than leading closed-source models while maintaining competitive quality.
- ◆ **Pinterest** said open-source models cost less than 8% per transaction compared with proprietary models of a similar parameter size.
- ◆ **HPE** uses open-weight model inferencing and intelligent routing in Private Cloud AI, claiming 30-60% lower costs than public-cloud token-based models.

Flexibility and control

- ◆ **Pinterest** can securely post-train open-source models in-house and swap or test multiple models side-by-side, including US-based models such as Gemma and models from Asia.
- ◆ **DeepInfra** offers shared serverless hosting and dedicated private deployments, allowing customers to run custom models or select specific performance profiles.
- ◆ Open models support data sovereignty and deployment flexibility, particularly for enterprise, government, and regulated customers.

Adoption and ecosystem benefits

- ◆ **NVIDIA** is developing open-source models such as NeMo as ecosystem enablers rather than direct threats to customers. Management said open source is important for:
 1. Broadening AI adoption.
 2. Ensuring access to high-performance models.
 3. Strengthening the US AI ecosystem.

AI monetization still in nascent stages

The meetings broadly suggested that AI monetization is moving beyond experimentation, although the revenue contribution is still modest relative to the underlying usage and investment.

- ◆ **Twilio was the clearest voice of caution:** AI-driven voice and agent adoption is promising, but monetization remains early. Customers often begin with limited production pilots, especially in regulated industries. AI has not yet materially changed adoption or revenue mix, and the pace of conversion from pilots to large-scale usage remains uncertain.
- ◆ **Pinterest sees strong engagement but monetization still has to catch up:** The platform has 640 million monthly active users and sustained user growth, but international monetization remains below peers. AI-related GPU investment of USD50m is a margin headwind, even though the business remains highly cash generative.
- ◆ **Adobe** has an AI-first ARR of approximately USD650m, still less than 2% of its roughly USD28bn revenue base. It is moving from seat-based subscriptions towards a hybrid model combining subscriptions with generative credits, allowing revenue to rise with usage.
- ◆ **HP Enterprise** is monetizing AI through integrated infrastructure and private-cloud deployments rather than token-based public-cloud consumption.
- ◆ **Cadence** is still in the “price discovery” phase for AI design tools. It has more than 20 customer engagements for agentic AI offerings, but monetization is not yet mature.

Strong earnings performance key for re-rating

AI-led demand is expected to remain robust, although growth will continue to be constrained by available supply capacity. While capital expenditure momentum among hyperscalers remains a critical driver of growth for semiconductor companies, the increasing adoption of artificial intelligence by enterprises and sovereign entities is becoming an increasingly important source of support.

This broader adoption is expected to benefit a wider range of companies across the semiconductor and technology ecosystems, extending beyond the largest hyperscalers. Against this backdrop, we expect valuation multiples for semiconductor and technology companies to remain capped in the short-term. Further re-rating is therefore likely to depend primarily on company-specific factors, particularly the delivery of strong earnings performance and the ability to exceed market expectations.

Companies mentioned in this report

Company	Ticker	Currency	CMP	TP	Rating
TSMC	2330 TT	TWD	2,480.00	3,400.00	Buy
Applied Materials	AMAT US	USD	464.24	641.00	Buy
Pinterest	PINS US	USD	19.36	23.40	Hold
Twilio	TWLO US	USD	266.06	211.00	Hold
Adobe	ADBE US	USD	249.52	310.00	Buy
SoftBank	9984 JP	JPY	6,247.00	NR	NR
Arista Networks	ANET US	USD	205.42	NR	NR
Lumentum	LITE US	USD	954.49	NR	NR
Cohu	COHU US	USD	60.62	NR	NR
Seagate	STX US	USD	877.33	NR	NR
Western Digital	WDC US	USD	448.17	NR	NR
HP Enterprise	HPE US	USD	61.75	NR	NR
Monolithic Power Systems	MPWR US	USD	1277.68	NR	NR
Cadence	CDNS US	USD	294.9	NR	NR
Lightmatter	Not listed				
DeepInfra	Not listed				
Celesta Capital	Not listed				

Source: Bloomberg, HSBC estimates

Priced as of close of 21 Sep 2026

Valuation and risks

Valuation		Risks
NVIDIA NVDA US	Current price: USD227.38	Downside risks: 1) Slower-than-expected mix shift towards higher ASP AI GPUs; 2) Lower investment from CSPs towards generative AI; 3) Weaker-than-expected overall data center momentum; and 4) Geopolitical uncertainty
	Target price: USD365.00 Up/downside: +60.5%	
Buy		
Methodology: FY28e target P/E multiple Assumptions: Target FY28e P/E of 25x applied to FY28e EPS of USD14.56. Our 25x target P/E is at a 15% premium to 1SD below the five-year historical average (21x). Given the recent rise of agentic AI, stocks exposed to server CPUs are trading at a higher premium and NVIDIA, which is mainly a GPU play, has traded at an average of 19x through the year. Therefore, we take 1SD above five-year historical average (21x) as the base. We apply a 15% premium, as we believe NVIDIA has upside to current earnings estimates; better supply capacity procurement vs peers; and a new narrative emerging with its increasing focus on open-source models. Hence, we believe our 25x target P/E is justified. Our TP and all our assumptions are unchanged. Frank Lee* frank.lee@hsbc.com.hk +852 2996 6916		
Intel Corp INTC US	Current price: USD121.78	Downside risks: 1) Worse-than-expected server CPU unit shipments; 2) weaker-than-expected server CPU pricing hike; and 3) higher margin dilution due to foundry ramp.
	Target price: USD200.00 Up/downside: +64.2%	
Buy		
Methodology: P/E-based SOTP Assumptions: Target FY27e P/E multiple of 30x (based on 2SD above the historical average P/E) applied to our core business 2027e EPS of USD4.25 implying core business value of USD125. Target FY27e P/B of 3.1x (in line with second tier foundries) applied to foundry BVPS of USD24.42 implying foundry business value of USD75.00. Our TP and all our assumptions are unchanged. Frank Lee* frank.lee@hsbc.com.hk +852 2996 6916		
Marvell MRVL US	Current price: USD257.38	Downside risks: 1) Slower-than-expected growth of optical transceivers market; 2) Higher-than-expected market share loss in DSPs; and 3) Delays in custom silicon projects
	Target price: USD325.00 Up/downside: +26.3%	
Buy		
Methodology: FY28e target P/E multiple. Assumptions: Target FY28e P/E of 42x applied to FY28e EPS of USD7.70. Our 42x FY28e target P/E multiple is the multiple between 2SD above the three-year historical average and the peak P/E multiple (52x). Marvell trades at c33x FY28e P/E based on our EPS estimates, between the three-year historical average and 1SD above three-year historical average. We believe the optical interconnect revenue opportunity is still severely underestimated by the market and there could be significant earnings upside surprise driven by stronger-than-expected optical interconnect revenue growth. In addition, we think CXL demand is also underestimated, hence we expect Marvell to re-rate higher than 2SD above the 3-year historical average P/E and towards the peak P/E multiple. Our USD325 TP implies c26% upside; accordingly, we have a Buy rating on the stock. Our TP and all our assumptions are unchanged. Frank Lee* frank.lee@hsbc.com.hk +852 2996 6916		
Qualcomm QCOM US	Current price: USD194.23	Upside risks: 1) Better-than-expected data center segment performance; 2) smartphone replacement cycle driven by AI smartphones; and 3) higher-than-expected premium handset mix Downside risks: 1) Weaker-than-expected smartphone demand; 2) slowdown in autos demand; and 3) geopolitical risks
	Target price: USD165.00 Up/downside: -15.0%	
Hold		
Methodology: FY27e target P/E multiple. Assumptions: Target P/E multiple of 16.3x applied to FY27e EPS of USD10.10. Our target P/E multiple of 16.3x is 1.25SD above the 5-year historical average. Given the aggressive data center ambitions, we believe the market is willing to give it a higher premium now. Our TP and all our assumptions are unchanged. Frank Lee* frank.lee@hsbc.com.hk +852 2996 6916		

GlobalFoundries GFS US	Current price: USD48.57 Target price: USD55.00 Up/downside: +13.2%	Valuation	Risks
Hold		Methodology: We value the stock on a target P/B multiple. Assumptions: Target P/B multiple of 2.6x applied to 2027e BVPS of USD21.78. Our target P/B is higher than the 5-year historical average of 2.3x due to improving supply-demand mechanics of mature foundry. We maintain our Hold rating as we expect a delay in ASP realization. Our TP and all our assumptions are unchanged. Frank Lee* frank.lee@hsbc.com.hk +852 2996 6916	Upside risks: 1) faster improvement in economies of scale at fabs than expected; 2) faster margin improvement as a result of stronger pricing upside and better cost management than anticipated; and 3) higher government subsidies than expected. Downside risks: 1) lack of economies of scale and increasing variable costs; 2) pricing power capped by high exposure to LTAs; and 3) increasing competition from mainland China, arising from any policy change.

Priced at close of 21 Sep 2026

 *Employed by a non-US affiliate of HSBC Securities (USA) Inc., and is not registered/ qualified pursuant to FINRA regulations
 Source: HSBC estimates

Disclosure appendix

Analyst Certification

The following analyst(s), economist(s), or strategist(s) who is(are) primarily responsible for this report, including any analyst(s) whose name(s) appear(s) as author of an individual section or sections of the report and any analyst(s) named as the covering analyst(s) of a subsidiary company in a sum-of-the-parts valuation certifies(y) that the opinion(s) on the subject security(ies) or issuer(s), any views or forecasts expressed in the section(s) of which such individual(s) is(are) named as author(s), and any other views or forecasts expressed herein, including any views expressed on the back page of the research report, accurately reflect their personal view(s) and that no part of their compensation was, is or will be directly or indirectly related to the specific recommendation(s) or views contained in this research report: Frank Lee and Pulkit Aggarwal

Important disclosures

Equities: Stock ratings and basis for financial analysis

HSBC and its affiliates, including the issuer of this report ("HSBC") believes an investor's decision to buy or sell a stock should depend on individual circumstances such as the investor's existing holdings, risk tolerance and other considerations and that investors utilise various disciplines and investment horizons when making investment decisions. Ratings should not be used or relied on in isolation as investment advice. Different securities firms use a variety of ratings terms as well as different rating systems to describe their recommendations and therefore investors should carefully read the definitions of the ratings used in each research report. Further, investors should carefully read the entire research report and not infer its contents from the rating because research reports contain more complete information concerning the analysts' views and the basis for the rating.

HSBC assigns ratings on the following basis:

The target price is based on the analyst's assessment of the stock's actual current value, although we expect it to take six to 12 months for the market price to reflect this. When the target price is more than 20% above the current share price, the stock will be classified as a Buy; when it is between 5% and 20% above the current share price, the stock may be classified as a Buy or a Hold; when it is between 5% below and 5% above the current share price, the stock will be classified as a Hold; when it is between 5% and 20% below the current share price, the stock may be classified as a Hold or a Reduce; and when it is more than 20% below the current share price, the stock will be classified as a Reduce.

Our ratings are re-calibrated against these bands at the time of any 'material change' (initiation or resumption of coverage, change in target price or estimates).

Upside/Downside is the percentage difference between the target price and the share price.

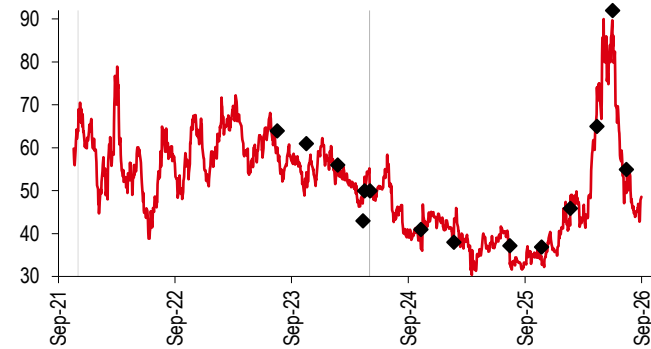
Rating distribution for long-term investment opportunities

As of 30 June 2026, the distribution of all independent ratings published by HSBC is as follows:

Buy	61%	(13% of these provided with Investment Banking Services in the past 12 months)
Hold	34%	(10% of these provided with Investment Banking Services in the past 12 months)
Sell	5%	(8% of these provided with Investment Banking Services in the past 12 months)

For the purposes of the distribution above the following mapping structure is used during the transition from the previous to current rating models: under our previous model, Overweight = Buy, Neutral = Hold and Underweight = Sell; under our current model Buy = Buy, Hold = Hold and Reduce = Sell. For rating definitions under both models, please see "Stock ratings and basis for financial analysis" above.

For the distribution of non-independent ratings published by HSBC, please see the disclosure page available at <http://www.hsbcnet.com/gbm/financial-regulation/investment-recommendations-disclosures>.

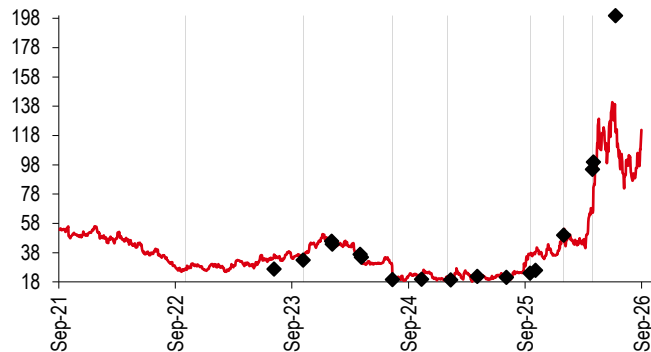
Share price and rating changes for long-term investment opportunities
**GlobalFoundries Inc (GFS.O) share price performance
USD Vs HSBC rating history**


Source: HSBC

Rating & target price history

From	To	Date	Analyst
N/A	Hold	22 Nov 2021	Frank Lee
Hold	Restricted	22 May 2024	
Restricted	Hold	24 May 2024	Frank Lee
Target price	Value	Date	Analyst
Price 1	64.00	08 Aug 2023	Frank Lee
Price 2	61.00	07 Nov 2023	Frank Lee
Price 3	56.00	13 Feb 2024	Frank Lee
Price 4	43.00	02 May 2024	Frank Lee
Price 5	50.00	07 May 2024	Frank Lee
Price 6	Restricted	22 May 2024	
Price 7	50.00	24 May 2024	Frank Lee
Price 8	41.00	31 Oct 2024	Frank Lee
Price 9	38.00	11 Feb 2025	Frank Lee
Price 10	37.21	05 Aug 2025	Frank Lee
Price 11	36.94	12 Nov 2025	Frank Lee
Price 12	45.93	11 Feb 2026	Frank Lee
Price 13	65.03	05 May 2026	Frank Lee
Price 14	92.00	23 Jun 2026	Frank Lee
Price 15	55.00	05 Aug 2026	Frank Lee

Source: HSBC

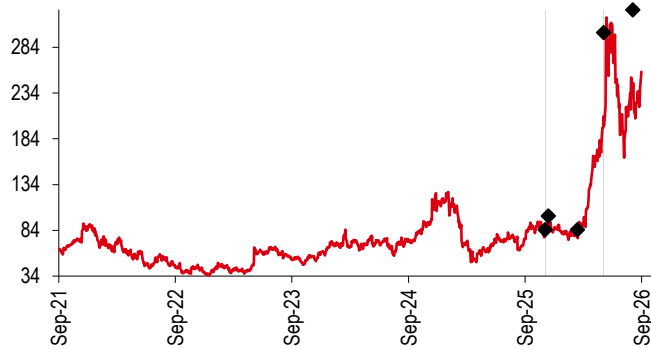
**Intel Corp (INTC.OQ) share price performance USD Vs
HSBC rating history**


Source: HSBC

Rating & target price history

From	To	Date	Analyst
N/A	Reduce	23 Oct 2022	Frank Lee
Reduce	Hold	27 Oct 2023	Frank Lee
Hold	Reduce	02 Aug 2024	Frank Lee
Reduce	Hold	21 Jan 2025	Frank Lee
Hold	Reduce	07 Oct 2025	Frank Lee
Reduce	Hold	20 Jan 2026	Frank Lee
Hold	Buy	21 Apr 2026	Frank Lee
Target price	Value	Date	Analyst
Price 1	27.00	28 Jul 2023	Frank Lee
Price 2	33.00	27 Oct 2023	Frank Lee
Price 3	46.00	24 Jan 2024	Frank Lee
Price 4	44.00	26 Jan 2024	Frank Lee
Price 5	37.00	22 Apr 2024	Frank Lee
Price 6	35.00	26 Apr 2024	Frank Lee
Price 7	19.80	02 Aug 2024	Frank Lee
Price 8	20.00	01 Nov 2024	Frank Lee
Price 9	19.50	31 Jan 2025	Frank Lee
Price 10	22.00	25 Apr 2025	Frank Lee
Price 11	21.25	25 Jul 2025	Frank Lee
Price 12	24.00	07 Oct 2025	Frank Lee
Price 13	26.00	24 Oct 2025	Frank Lee
Price 14	50.00	20 Jan 2026	Frank Lee
Price 15	95.00	21 Apr 2026	Frank Lee
Price 16	100.00	24 Apr 2026	Frank Lee
Price 17	200.00	02 Jul 2026	Frank Lee

Source: HSBC

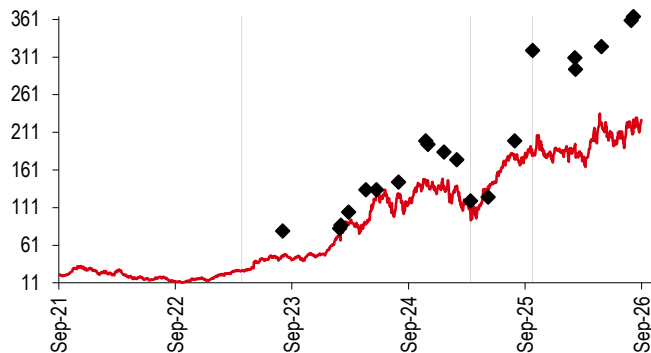
**Marvell Technology (MRVL.N) share price performance
USD Vs HSBC rating history**


Source: HSBC

Rating & target price history

From	To	Date	Analyst
N/A	Hold	23 Nov 2025	Frank Lee
Hold	Buy	26 May 2026	Frank Lee
Target price	Value	Date	Analyst
Price 1	85.00	23 Nov 2025	Frank Lee
Price 2	100.00	03 Dec 2025	Frank Lee
Price 3	85.00	04 Mar 2026	Frank Lee
Price 4	300.00	26 May 2026	Frank Lee
Price 5	325.00	25 Aug 2026	Frank Lee

Source: HSBC

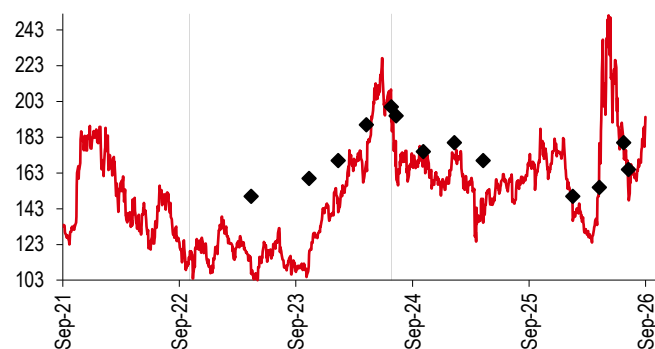
**NVIDIA Corp (NVDA.OQ) share price performance USD
Vs HSBC rating history**


Source: HSBC

Rating & target price history

From	To	Date	Analyst
Reduce	Buy	17 Apr 2023	Frank Lee
Buy	Hold	03 Apr 2025	Frank Lee
Hold	Buy	15 Oct 2025	Frank Lee
Target price	Value	Date	Analyst
Price 1	80.00	24 Aug 2023	Frank Lee
Price 2	83.50	18 Feb 2024	Frank Lee
Price 3	88.00	22 Feb 2024	Frank Lee
Price 4	105.00	17 Mar 2024	Frank Lee
Price 5	135.00	10 May 2024	Frank Lee
Price 6	135.00	13 Jun 2024	Frank Lee
Price 7	145.00	21 Aug 2024	Frank Lee
Price 8	200.00	14 Nov 2024	Frank Lee
Price 9	195.00	21 Nov 2024	Frank Lee
Price 10	185.00	10 Jan 2025	Frank Lee
Price 11	175.00	19 Feb 2025	Frank Lee
Price 12	120.00	03 Apr 2025	Frank Lee
Price 13	125.00	29 May 2025	Frank Lee
Price 14	200.00	20 Aug 2025	Frank Lee
Price 15	320.00	15 Oct 2025	Frank Lee
Price 16	310.00	24 Feb 2026	Frank Lee
Price 17	295.00	26 Feb 2026	Frank Lee
Price 18	325.00	19 May 2026	Frank Lee
Price 19	360.00	20 Aug 2026	Frank Lee
Price 20	365.00	27 Aug 2026	Frank Lee

Source: HSBC

Qualcomm (QCOM.OQ) share price performance USD Vs HSBC rating history

Rating & target price history

From	To	Date	Analyst
N/A	Buy	23 Oct 2022	Frank Lee
Buy	Hold	17 Jul 2024	Frank Lee
Target price	Value	Date	Analyst
Price 1	150.00	04 May 2023	Frank Lee
Price 2	160.00	02 Nov 2023	Frank Lee
Price 3	170.00	01 Feb 2024	Frank Lee
Price 4	190.00	29 Apr 2024	Frank Lee
Price 5	200.00	17 Jul 2024	Frank Lee
Price 6	195.00	01 Aug 2024	Frank Lee
Price 7	175.00	25 Oct 2024	Frank Lee
Price 8	180.00	30 Jan 2025	Frank Lee
Price 9	170.00	01 May 2025	Frank Lee
Price 10	150.00	05 Feb 2026	Frank Lee
Price 11	155.00	30 Apr 2026	Frank Lee
Price 12	180.00	15 Jul 2026	Frank Lee
Price 13	165.00	30 Jul 2026	Frank Lee

Source: HSBC

To view a list of all the independent fundamental ratings/recommendations disseminated by HSBC during the preceding 12-month period, and the location where we publish our quarterly distribution of non-fundamental recommendations (applicable to Fixed Income and Currencies research only), please use the following links to access the disclosure page:

Clients of HSBC Private Bank: www.research.privatebank.hsbc.com/Disclosures

All other clients: www.research.hsbc.com/A/Disclosures

HSBC & Analyst disclosures
Disclosure checklist

Company	Ticker	Recent price	Price date	Disclosure
GLOBALFOUNDRIES INC	GFS.O	48.57	21 Sep 2026	7
INTEL CORP	INTC.OQ	121.78	21 Sep 2026	6, 7
MARVELL TECHNOLOGY	MRVL.N	257.38	21 Sep 2026	1, 4, 5, 7
NVIDIA CORP	NVDA.OQ	227.38	21 Sep 2026	1, 5, 7
QUALCOMM	QCOM.OQ	194.23	21 Sep 2026	7

Source: HSBC

- 1 HSBC has managed or co-managed a public offering of securities for this company within the past 12 months.
- 2 HSBC expects to receive or intends to seek compensation for investment banking services from this company in the next 3 months.
- 3 At the time of publication of this report, HSBC Securities (USA) Inc. is a Market Maker in securities issued by this company.
- 4 As of 31 August 2026, HSBC beneficially owned 1% or more of a class of common equity securities of this company.
- 5 This company was a client of HSBC or had during the preceding 12 month period been a client of and/or paid compensation to HSBC in respect of investment banking services.
- 6 This company was a client of HSBC or had during the preceding 12 month period been a client of and/or paid compensation to HSBC in respect of non-investment banking securities-related services.
- 7 This company was a client of HSBC or had during the preceding 12 month period been a client of and/or paid compensation to HSBC in respect of non-securities services.
- 8 A covering analyst/s has received compensation from this company in the past 12 months.
- 9 A covering analyst/s or a member of his/her household has a financial interest in the securities of this company, as detailed below.
- 10 A covering analyst/s or a member of his/her household is an officer, director or supervisory board member of this company, as detailed below.
- 11 At the time of publication of this report, HSBC is a non-US Market Maker in securities issued by this company and/or in securities in respect of this company.
- 12 As of 16 September 2026, HSBC beneficially held a net long position of more than 0.5% of this company's total issued share capital, calculated according to the SSR methodology.

13 As of 16 September 2026, HSBC beneficially held a net short position of more than 0.5% of this company's total issued share capital, calculated according to the SSR methodology.

14 HSBC Qianhai Securities Limited holds 1% or more of a class of common equity securities of this company.

HSBC and its affiliates will from time to time sell to and buy from customers the securities/instruments, both equity and debt (including derivatives) of companies covered in HSBC Research on a principal or agency basis or act as a market maker or liquidity provider in the securities/instruments mentioned in this report.

Analysts, economists, and strategists are paid in part by reference to the profitability of HSBC which includes investment banking, sales & trading, and principal trading revenues.

Whether, or in what time frame, an update of this analysis will be published is not determined in advance.

Non-U.S. analysts may not be associated persons of HSBC Securities (USA) Inc, and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with the subject company, public appearances and trading securities held by the analysts.

Economic sanctions laws imposed by certain jurisdictions such as the US, the EU, the UK, and others, may prohibit persons subject to those laws from making certain types of investments, including by transacting or dealing in securities of particular issuers, sectors, or regions. This report does not constitute advice in relation to any such laws and should not be construed as an inducement to transact in securities in breach of such laws.

For disclosures in respect of any company mentioned in this report, please see the most recently published report on that company available at www.hsbcnet.com/research. HSBC Private Bank clients should contact their Relationship Manager for queries regarding other research reports. In order to find out more about the proprietary models used to produce this report, please contact the authoring analyst.

HSBC may use Artificial Intelligence (AI) tools approved for adoption within HSBC in the development of its research reports, utilizing these technologies to analyse large volumes of data, enhance efficiency and improve the overall user experience. This includes but is not limited to paraphrasing, developing suitable captions and supporting data visualisation. It is important to note that while AI tools assist in various aspects of report creation, all investment recommendations and opinions presented herein are formulated and approved exclusively by our research analysts. The final content of the report reflects the professional judgement and expertise of our research analysts, ensuring compliance with regulatory standards and maintaining the integrity of our research process.

Additional disclosures

- 1 This report is dated as at 23 September 2026.
- 2 All market data included in this report are dated as at close 21 September 2026, unless a different date and/or a specific time of day is indicated in the report.
- 3 HSBC has procedures in place to identify and manage any potential conflicts of interest that arise in connection with its Research business. HSBC's analysts and its other staff who are involved in the preparation and dissemination of Research operate and have a management reporting line independent of HSBC's Investment Banking business. Information Barrier procedures are in place between the Investment Banking, Principal Trading, and Research businesses to ensure that any confidential and/or price sensitive information is handled in an appropriate manner.
- 4 You are not permitted to use, for reference, any data in this document for the purpose of (i) determining the interest payable, or other sums due, under loan agreements or under other financial contracts or instruments, (ii) determining the price at which a financial instrument may be bought or sold or traded or redeemed, or the value of a financial instrument, and/or (iii) measuring the performance of a financial instrument or of an investment fund.
- 5 As of 16 September 2026 HSBC owned a significant interest in the debt securities of the following company(ies): INTEL CORP, NVIDIA CORP, QUALCOMM

Production & distribution disclosures

1. This report was produced and signed off by the author on 22 Sep 2026 09:28 GMT.
2. In order to see when this report was first disseminated please see the disclosure page available at <https://www.research.hsbc.com/R/34/lwkSfnk>

Disclaimer

Legal entities as at 13 May 2026:

HSBC Bank plc; HSBC Continental Europe; HSBC Continental Europe SA, Germany; HSBC Bank Middle East Limited, DIFC; HSBC Bank Middle East Limited, Dubai branch; HSBC Yatirim Menkul Degerler AS, Istanbul; The Hongkong and Shanghai Banking Corporation Limited, Hong Kong; The Hongkong and Shanghai Banking Corporation Limited, Singapore Branch; The Hongkong and Shanghai Banking Corporation Limited, Seoul Securities Branch; The Hongkong and Shanghai Banking Corporation Limited, Seoul Branch; HSBC Qianhai Securities Limited; HSBC Securities (Taiwan) Corporation Limited; HSBC Securities and Capital Markets (India) Private Limited, Mumbai; HSBC Bank Australia Limited; HSBC Securities (USA) Inc., New York; HSBC México, SA, Institución de Banca Múltiple, Grupo Financiero HSBC; Banco HSBC SA

Issuer of report

The Hongkong and Shanghai Banking Corporation Limited
 Level 16, 1 Queen's Road Central
 Hong Kong SAR
 Telephone: +852 2843 9111
 Fax: +852 2596 0200
 Website: www.research.hsbc.com

This document has been issued by The Hongkong and Shanghai Banking Corporation Limited ("HSBC") in the conduct of its Hong Kong regulated business for the information of its institutional and professional investor (as defined by Securities and Future Ordinance (Chapter 571)) customers; it is not intended for and should not be distributed to retail customers in Hong Kong, unless permitted otherwise. The Hongkong and Shanghai Banking Corporation Limited is regulated by the Hong Kong Monetary Authority. All enquires by recipients in Hong Kong must be directed to your HSBC contact in Hong Kong. If it is received by a customer of an affiliate of HSBC, its provision to the recipient is subject to the terms of business in place between the recipient and such affiliate. This document is not and should not be construed as an offer to sell or the solicitation of an offer to purchase or subscribe for any investment. HSBC has based this document on information obtained from sources it believes to be reliable but which it has not independently verified; HSBC makes no guarantee, representation or warranty and accepts no responsibility or liability as to its accuracy or completeness. Expressions of opinion are those of the Research Division of HSBC only and are subject to change without notice. From time to time research analysts conduct site visits of covered issuers. HSBC policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits. HSBC and its affiliates and/or their officers, directors and employees may have positions in any securities mentioned in this document (or in any related investment) and may from time to time add to or dispose of any such securities (or investment). HSBC and its affiliates may act as market maker or have assumed an underwriting commitment in the securities of companies discussed in this document (or in related investments), may sell them to or buy them from customers on a principal basis and may also perform or seek to perform investment banking or underwriting services for or relating to those companies. The document is intended to be distributed in its entirety. Unless governing law permits otherwise, you must contact a HSBC Group member in your home jurisdiction if you wish to use HSBC Group services in effecting a transaction in any investment mentioned in this document.

In the UK, this publication is distributed by HSBC Bank plc for the information of its Clients (as defined in the Rules of FCA) and those of its affiliates only. Nothing herein excludes or restricts any duty or liability to a customer which HSBC Bank plc has under the Financial Services and Markets Act 2000 or under the Rules of FCA and PRA. A recipient who chooses to deal with any person who is not a representative of HSBC Bank plc in the UK will not enjoy the protections afforded by the UK regulatory regime. HSBC Bank plc is regulated by the Financial Conduct Authority and the Prudential Regulation Authority.

In the European Economic Area, this publication has been distributed by HSBC Continental Europe or by such other HSBC affiliate from which the recipient receives relevant services.

In Japan, this publication has been distributed by HSBC Securities (Japan) Co., Ltd.. It may not be further distributed in whole or in part for any purpose. In Korea, this publication is distributed by either The Hongkong and Shanghai Banking Corporation Limited, Seoul Securities Branch ("HBAP SLS") or The Hongkong and Shanghai Banking Corporation Limited, Seoul Branch ("HBAP SEL") for the general information of professional investors specified in Article 9 of the Financial Investment Services and Capital Markets Act ("FSCMA"). This publication is not a prospectus as defined in the FSCMA. It may not be further distributed in whole or in part for any purpose. Both HBAP SLS and HBAP SEL are regulated by the Financial Services Commission and the Financial Supervisory Service of Korea. In Singapore, this publication is distributed by The Hongkong and Shanghai Banking Corporation Limited, Singapore Branch for the general information of institutional investors or other persons specified in Sections 274 and 304 of the Securities and Futures Act 2001 of Singapore ("SFA") and accredited investors and other persons in accordance with the conditions specified in Sections 275 and 305 of the SFA. Only Economics or Currencies reports are intended for distribution to a person who is not an Accredited Investor, Expert Investor or Institutional Investor as defined in SFA. The Hongkong and Shanghai Banking Corporation Limited, Singapore Branch accepts legal responsibility for the contents of reports. This publication is not a prospectus as defined in the SFA. It may not be further distributed in whole or in part for any purpose. The Hongkong and Shanghai Banking Corporation Limited Singapore Branch is regulated by the Monetary Authority of Singapore. Recipients in Singapore should contact a "Hongkong and Shanghai Banking Corporation Limited, Singapore Branch" representative in respect of any matters arising from, or in connection with this report. Please refer to The Hongkong and Shanghai Banking Corporation Limited Singapore Branch's website at www.business.hsbc.com.sg for contact details. In Australia, this publication has been distributed by The Hongkong and Shanghai Banking Corporation Limited (ABN 65 117 925 970, AFSL 301737) for the general information of its "wholesale" customers (as defined in the Corporations Act 2001). Where distributed to retail customers, this research is distributed by HSBC Bank Australia Limited (ABN 48 006 434 162, AFSL No. 232595). These respective entities make no representations that the products or services mentioned in this document are available to persons in Australia or are necessarily suitable for any particular person or appropriate in accordance with local law. No consideration has been given to the particular investment objectives, financial situation or particular needs of any recipient.

HSBC Securities (USA) Inc. accepts responsibility for the content of this research report prepared by its non-US foreign affiliate. The information contained herein is under no circumstances to be construed as investment advice and is not tailored to the needs of the recipient. All US persons receiving and/or accessing this report and wishing to effect transactions in any security discussed herein should do so with HSBC Securities (USA) Inc. in the United States and not with its non-US foreign affiliate, the issuer of this report. HSBC México, S.A., Institución de Banca Múltiple, Grupo Financiero HSBC is authorized and regulated by Secretaría de Hacienda y Crédito Público and Comisión Nacional Bancaria y de Valores (CNBV). In Brazil, this document has been distributed by Banco HSBC SA ("HSBC Brazil"), and/or its affiliates. As required by Resolution No. 20/2021 of the Securities and Exchange Commission of Brazil (Comissão de Valores Mobiliários), potential conflicts of interest concerning (i) HSBC Brazil and/or its affiliates; and (ii) the analyst(s) responsible for authoring this report are stated on the chart above labelled "HSBC & Analyst Disclosures". If you are a customer of HSBC International Wealth & Premier Banking ("IWPB"), including HSBC Private Bank, you are eligible to receive this publication only if: (i) you have been approved to receive relevant research publications by an applicable HSBC legal entity; (ii) you have agreed to the applicable HSBC entity's terms and conditions and/or customer declaration for accessing research; and (iii) you have agreed to the terms and conditions of any other internet banking, online banking, mobile banking and/or investment services offered by that HSBC entity, through which you will access research publications (collectively with (ii), the "Terms"). If you do not meet the above eligibility requirements, please disregard this publication and, if you are a IWPB customer, please notify your Relationship Manager or call the relevant customer hotline. Distribution of this publication is the sole responsibility of the HSBC entity with whom you have agreed the Terms. Receipt of research publications is strictly subject to the Terms and any other conditions or disclaimers applicable to the provision of the publications that may be advised by IWPB.

Should you wish to use AI technology (computational, statistical, machine-learning or other artificial intelligence techniques to infer or learn from data to find patterns, take actions, make decisions, or generate output) to help you analyse, summarise or evaluate this publication and/or any other research materials provided to you by HSBC, you shall seek HSBC consent in advance of any such use and not upload this publication and/or any other research materials produced by HSBC to an AI system or otherwise use AI technology in connection with the research services HSBC provides to you without our consent.

© Copyright 2026, The Hongkong and Shanghai Banking Corporation Limited, ALL RIGHTS RESERVED. No part of this publication may be reproduced, stored in a retrieval system, or transmitted, on any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the prior written permission of The Hongkong and Shanghai Banking Corporation Limited.

[1290208]